Apeksha Jain

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Education

ABV-Indian Institute of Information Technology and Management, Gwalior Aug 2023 – Jun 2028
Integrated B.Tech + M.Tech in Information Technology
Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Board of Secondary Education, Madhya Pradesh Apr 2022 – Apr 2023
Class XII *Percentage: 91%*

Projects

PayLite – UPI-Style Digital Payments Wallet *React · TypeScript · Node.js · Express · MongoDB · Razorpay*
- Engineered a peer-to-peer payments wallet with unique PayLite IDs, QR scan-to-pay and collect requests, built on a double-entry ledger so every transaction balances.
- Guaranteed exactly-once, overdraft-free transfers by storing idempotency keys in the same MongoDB transaction as the payment with conditional debits; 10 concurrent payments never overdraw a wallet.
- Integrated Razorpay top-ups with server-side HMAC signature verification and a signed webhook backup, so each payment is credited exactly once even when both confirmations arrive together.
- Added fraud-style safeguards (PIN lockout, daily transfer limits), spending insights via MongoDB aggregation, and 37 automated CI tests.

Imagify – AI Image Generation SaaS with Payments *React · Node.js · Express · MongoDB · Razorpay*
- Built a full-stack text-to-image web app on Cloudflare Workers AI (FLUX.1) with a ClipDrop fallback and a paid credit system.
- Secured Razorpay checkout with server-set pricing, HMAC-SHA256 verification and replay-safe atomic credit updates.
- Prevented credit overspending under concurrent requests with a single atomic MongoDB update per request, with automatic refunds when generation fails.
- Hardened the REST API with JWT + bcrypt authentication, rate limiting and input validation; tested concurrency with Supertest.

Revela AI – Adaptive Interview Platform *Next.js · TypeScript · REST APIs · MongoDB*
- Built the Next.js 15 frontend of an adaptive AI interview platform covering candidates, interview sessions, reports and analytics.
- Designed typed TypeScript service layers that consume the backend's REST web services across every dashboard view.
- Implemented OAuth and email/password sign-in with NextAuth.js v5 and protected routes via Next.js middleware.
- Built a responsive Tailwind CSS interface; collaborated with the team through Git and code reviews, deploying on Vercel.

Research

Undergraduate Researcher – ABV-IIITM Gwalior *May 2026 – Present*
ICOT-RAG: Explainable IoT Cybersecurity QA | Python, RAG, ChromaDB, LLMs *Guide: Dr. Bhanu Chander*
- Built an iterative, facet-aware RAG pipeline in Python that retrieves evidence across IoT attack behaviors, techniques, vulnerabilities, exploits and mitigations.
- Integrated 1,837+ documents from IoT-23, MITRE ATT&CK and VARIoT into ChromaDB, raising facet recall from 0.69 to 1.00 on a 50-question benchmark.
- First-author manuscript under revision for ACM DTRAP.

Technical Skills

Programming Languages: JavaScript, TypeScript, Java, C++, Python, SQL
Web Technologies: HTML5, CSS3, React.js, Next.js, Tailwind CSS, HTTP, DOM, AJAX
Backend & APIs: Node.js, Express.js, REST APIs, Web Services, JWT Authentication, Payment Gateway Integration
Databases: MongoDB
Core Computer Science: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Developer Tools: Git, GitHub, Postman, VS Code, Vercel, Render

Achievements & Certifications

AWS Certified AI Practitioner – Amazon Web Services.
ICPC Algo Queen 2026 – Rank 187 (College), ICPC-endorsed Girls' Programming Cup by Amrita Vishwa Vidyapeetham.
Google BigCode – Won the Agent AI Track and advanced to the semi-finals.
Hackathons – Finalist at Hackvega 2.0; Top 5 at Xenith Hackathon (IEEE JIIT).
LeetCode 365-Day Badge – Maintained a year-long daily problem-solving streak.
Programs – SheFi Scholar (2024–25); Codess Cafe Mentee; GirlScript Summer of Code and Flipkart GRiD participant.